INR Monitoring
(Personalized INR Monitoring & Warfarin Dose Optimization System)

Computer Engineering - Senior Project Final Report

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Summary

Warfarin is an oral anticoagulant used for the prevention and treatment of thromboembolic events in patient groups such as atrial fibrillation, stroke history, mechanical heart valve replacement, deep vein thrombosis, and pulmonary embolism. Although Warfarin is clinically effective, it requires careful and continuous monitoring through the International Normalized Ratio (INR). If INR is below the target therapeutic range, the risk of thrombosis increases; if INR is above the target range, the risk of bleeding increases. For this reason, Warfarin management is a sensitive clinical process that depends on regular INR measurements, dose adjustment, patient education, and awareness of food and drug interactions.

This project was developed to support patients who use Warfarin by providing a mobile system for INR monitoring, weekly dose planning, warning generation, and rule-based dose recommendation. The original idea was not only to create a simple INR diary but to design a clinically structured decision-support prototype that can take the patient's current INR value, current weekly Warfarin dose, and treatment indication as input, then produce an explainable weekly dose suggestion, daily tablet schedule, warning messages, and next INR control timing.

The system was implemented as a React Native and Expo mobile application using TypeScript. Firebase Authentication was used for user login and registration, and Cloud Firestore was used as the serverless backend database for profile data, INR records, treatment information, recommendations, and user preferences. Firestore Security Rules were designed so that users can only access their own data. The application includes a medical consent flow, login and registration, indication selection, profile management, INR entry, INR history, home dashboard, weekly dose visualization, medical warnings, food and drug interaction information, privacy policy, help/about

pages, light/dark theme support, language preference support, and account deletion with reauthentication.

The clinical side of the project was built around a rule-based Warfarin dose adjustment algorithm. Different target INR ranges were defined for atrial fibrillation/stroke, single mechanical valve, double mechanical valve, and thromboembolic indications. The algorithm applies percentage-based weekly dose increase or decrease rules such as +1/14, +1/7, -1/14, or -1/7, depending on the INR interval and indication. In high INR scenarios, the algorithm recommends temporary medication stop periods and short-term INR recheck timing. Warning codes U-1 to U-12 were designed to represent clinical safety messages such as possible need for injection therapy, bleeding risk, same-time daily medication use, informing healthcare professionals before procedures, and forgotten-dose behavior.

The project also included academic and release preparation work. A literature review was performed on Warfarin pharmacology, INR physiology, target ranges, dose adjustment protocols, guideline-based anticoagulation management, vitamin K interactions, drug interactions, home INR monitoring, and similar mobile health applications. The project was also prepared as a TUBITAK 2209-B research project, including problem definition, innovation, technical architecture, frontend/backend design, algorithm design, risk management, research plan, and commercialization potential. Figma screen designs, academic presentations, posters, demo materials, GitHub documentation, and Google Play preparation assets were produced. During the release process, Android package configuration, app icons, adaptive icons, splash screen, privacy policy, medical disclaimer, screenshots, feature graphics, Android App Bundle builds, APK builds, versionCode updates, and Google Play testing track requirements were handled.

At the end of the development process, the project evolved from a simple tracking idea into a research-based mobile health decision-support prototype. It combines clinical rules, mobile usability, backend data storage, security rules, release engineering, academic documentation, and future machine-learning planning in a single integrated system.

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Introduction

Warfarin therapy is one of the most well-known examples of a treatment process that requires continuous monitoring and individualized dose adjustment. Warfarin has a narrow therapeutic window, meaning that small changes in dose, diet, other medications, or patient condition can change the anticoagulation effect. The INR value is used to monitor the anticoagulant effect and to determine whether the patient is within the safe therapeutic range.

In daily life, many Warfarin patients, especially elderly patients, face difficulties in understanding their dose plan after each INR measurement. A physician may recommend a new weekly dose, temporary dose interruption, follow-up control date, or warning about bleeding/thrombosis risk. However, patients may have difficulty converting a weekly dose into a clear day-by-day tablet plan. They may also forget which foods and drugs interact with Warfarin, or may not know when a high INR value requires urgent medical attention.

The motivation of this project came from this practical clinical problem. The aim was to build a mobile system that helps the patient record INR values, understand whether the INR is below, inside, or above the therapeutic range, receive a structured rule-based recommendation, view a weekly tablet distribution, and see relevant warnings. The application was designed as an academic prototype and clinical decision-support concept. It does not replace physicians, medical diagnosis, or direct clinical supervision. Instead, it demonstrates how digital health software can organize clinical rules and patient information in a safer, more understandable, and more accessible format.

The project was implemented using modern mobile development technologies. React Native and Expo were selected for cross-platform mobile development, TypeScript was used for safer code structure, Firebase Authentication was used for account management, and Cloud Firestore was used for cloud data storage. The project also included a long design and iteration process, including Figma screens, UI refinements, medical disclaimer placement, profile flow, initial indication selection, home dashboard redesign, dark/light theme support, language support, and Google Play release configuration.

Literature Review and Background

Before implementation, a literature and domain review was performed to understand both the medical and technical dimensions of the problem. The medical research focused on Warfarin pharmacology, INR monitoring, therapeutic target ranges, dose adjustment protocols, anticoagulation safety, bleeding and thrombosis risk, vitamin K metabolism, drug interactions, food interactions, and patient self-monitoring. The technical research focused on mobile health applications, similar INR tracking applications, Firebase-based mobile backends, React Native development, secure user data storage, and app store release requirements.

Warfarin works by interfering with vitamin K dependent clotting factor synthesis. Because vitamin K intake, other medications, alcohol use, illness, and adherence can influence anticoagulation response, INR monitoring is essential. Many guidelines and clinical references describe target INR ranges such as 2.0-3.0 for many common indications and higher target ranges for selected mechanical valve contexts. This project used indication-based target ranges and rule-based adjustment intervals rather than a single universal threshold.

The literature review also showed that many existing mobile applications focus mainly on recording INR values. Some apps provide reminders or history logs, but they often do not include structured dose adjustment logic, clinical warning codes, weekly tablet distribution, or integrated food/drug interaction education. This created the main innovation area for this project: combining INR tracking with a transparent rule-based recommendation engine and patient-facing safety information.

Home INR monitoring and patient self-testing were also reviewed. These approaches show the importance of giving patients a clearer role in their own anticoagulation management, but the interpretation of INR values and dose changes must remain clinically cautious. For this reason, the application was designed with strong disclaimers and warning messages. It does not make autonomous medical decisions; instead, it presents rule-based guidance and encourages physician consultation for risky cases.

The technical literature and documentation review supported the use of React Native and Expo for mobile development, Firebase Authentication for user accounts, and Cloud Firestore for structured cloud storage. Google Play documentation was also reviewed during release preparation, especially internal, closed, and open testing tracks, Android App Bundle requirements, version code handling, and tester distribution workflows.

Problem Definition

The main problem addressed in this project is the difficulty of safe, understandable, and consistent Warfarin dose follow-up outside the clinic. Patients often receive INR results and dose instructions, but the instructions may be difficult to remember or convert into daily medication behavior. This problem becomes more important when the patient is elderly, has multiple medications, or has limited technical/medical literacy.

The project frames Warfarin dose adjustment as a constrained decision-support problem. The system receives the current INR value, current weekly Warfarin dose, and treatment indication. It then determines the target INR range, selects the related clinical rule interval, calculates the weekly dose change, rounds the weekly dose to a tablet-compatible value, generates warning codes, and presents the next INR control timing.

The problem has three major constraints. First, the output must be clinically conservative and explainable. Second, the interface must be simple enough for non-technical and older users. Third, the backend must protect sensitive health-related user data. Therefore, the final design combines rule-based calculation, clear visual presentation, medical disclaimers, and Firestore ownership rules.

Project Objectives

The main objective of the project is to develop a mobile application prototype that helps Warfarin patients monitor INR values and understand dose-related guidance in a structured way.

The specific objectives are:

- To create a mobile INR monitoring application using React Native, Expo, and TypeScript.
- To implement Firebase Authentication for secure login, registration, session handling, and account management.
- To store user profiles, treatment indications, INR records, theme/language preferences, and recommendation-related data in Cloud Firestore.
- To design a rule-based Warfarin dose recommendation engine based on INR value, weekly dose, and indication-specific target range.
- To generate weekly dose plans and day-by-day tablet schedules.
- To provide warning messages for low INR, high INR, bleeding risk, thrombosis risk, and urgent medical attention scenarios.
- To include food and drug interaction information related to Warfarin and vitamin K.
- To build a user interface suitable for elderly users, with simple navigation, readable cards, clear buttons, and visual tablet icons.
- To prepare the application for academic presentation, TUBITAK 2209-B documentation, GitHub documentation, and Google Play testing/release workflows.

Requirement Analysis

The functional requirements of the system were defined around the complete patient flow. A user must be able to accept a medical consent statement, create an account, log in, select the reason for Warfarin use, enter treatment profile information, enter INR measurements, receive a recommendation, view the latest INR result on the home screen, see the daily and weekly dose schedule, review INR history, and access medical warnings and interaction information.

The profile system stores treatment-related information such as indication, target INR minimum and maximum, current weekly dose, tablet strength, theme mode, and language. The INR entry system was designed carefully because decimal input can cause mistakes on mobile keyboards. For this reason, manual free-text INR entry was replaced with a picker-based input structure so that users can choose values in a controlled range, including very high INR values such as 8+ or 9+ depending on the design iteration.

The non-functional requirements include usability, readability, data privacy, explainability, maintainability, and release readiness. Since this is a health-related application, the system must show that it is an academic prototype and not a replacement for medical care. The user interface must be clear and consistent. The codebase must separate screens, services, algorithms, constants, navigation, theme, and types. The backend must ensure user data ownership, and the release configuration must preserve package name and versioning for Google Play uploads.

System Design and Architecture

The system architecture has three main layers: mobile frontend, clinical algorithm layer, and Firebase backend layer.

The frontend layer was built with React Native and Expo. It contains all user-facing screens, including Login, Medical Consent, Sign Up, Indication Reason, Home, Enter INR, INR History, User Profile, Settings, Privacy Policy, About, Medical Warnings, Food Interactions, Drug Interactions, Help, and Delete Account. React Navigation Native Stack manages screen transitions. React Context is used for theme and language state.

The algorithm layer is written in TypeScript under the algorithms module. The active mobile calculation path uses the `calculateDose()` function. This function receives the patient indication, current INR, and weekly dose, determines the target range, applies the correct protocol rule, calculates the new weekly dose, rounds it to a half-tablet-compatible 2.5 mg step, and returns the target range, suggested weekly dose, action text, warnings, and next check timing.

The backend layer uses Firebase rather than a custom Node.js or Express server. Firebase Authentication handles user identity and session state. Cloud Firestore stores user profiles, INR records, recommendation-related data, and preferences. The app communicates with Firebase directly through the Firebase SDK. Firestore Security Rules are used to restrict access so that each user can only read, create, update, or delete documents that belong to their own `uid`. This makes the backend serverless, scalable for a prototype, and easier to maintain within the project scope.

The repository was organized into separate folders for algorithms, components, constants, navigation, screens, services, theme, and types. This structure made it possible to separate clinical rules from UI code and backend service functions.

Clinical Algorithm Design

The dose recommendation algorithm is the core of the project. The first design decision was to use a rule-based system instead of machine learning. This was important because the project did not yet have a large validated patient dataset. In a medical decision-support context, explainability and safety are more important than complex prediction. A rule-based system allows every output to be traced back to a known clinical interval and action rule.

The algorithm accepts three main inputs:

- Current INR value.
- Current weekly Warfarin dose in milligrams.
- Treatment indication.

The supported indication categories include:

- Atrial fibrillation and stroke-related anticoagulation.
- Single mechanical heart valve.
- Double mechanical heart valve.
- Thromboembolic conditions such as deep vein thrombosis and pulmonary embolism.

For each indication, a target INR range is defined. The algorithm then checks which INR interval the current value belongs to. If the INR is below target, the weekly dose is increased by a controlled ratio such as +1/14 or +1/7. If the INR is inside the target range, the same dose is continued. If the INR is above target, the weekly dose is decreased by ratios such as -1/14 or -1/7. If the INR is very high, the output changes from a normal dose adjustment to a safety action, such as stopping the medication for 2, 3, or 4 days and repeating INR measurement after a short interval.

For example, in several protocols, an INR between 5.0 and 6.0 triggers “medication stop, wait 2 days” and recommends INR control on the third day. INR between 6.0 and 7.0 triggers a longer stop period and control on the fourth day. INR above 7.0 triggers a more serious stop recommendation and control on the fifth day. These rules are supported by warning codes such as U-4, U-5, and U-6.

The weekly dose result is rounded to a 2.5 mg step. This is important because the project assumes practical tablet use with 5 mg and half-tablet 2.5 mg units. The daily distribution module then converts the weekly total into a day-by-day schedule. The user sees both the weekly total and the daily tablet plan visually, reducing the cognitive burden of converting milligrams into medication behavior.

The algorithm also produces warning codes. The warning system includes twelve clinical warning categories, U-1 to U-12. These warnings cover situations such as possible need for injection treatment, low INR risk, high INR bleeding risk, life-threatening bleeding symptoms, taking the medication at the same time every day, informing healthcare professionals before procedures, and what to do if a dose is forgotten.

Frontend Mobile Application Development

The mobile application went through many design iterations. In the early version, the home screen mainly showed the INR result. Later, it evolved into a full dashboard that includes current INR, target range, today's dose, weekly dose plan, treatment recommendation, warnings, and next control timing.

The main frontend screens are:

- Login Screen: Allows existing users to sign in through Firebase Authentication.
- Medical Consent Screen: Shows a medical disclaimer before registration.
- Sign Up Screen: Creates a new user account.
- Indication Reason Screen: Collects the initial reason for Warfarin use.
- Home Screen: Displays current INR status, weekly dose, today's dose, warning messages, and navigation menu.
- Enter INR Screen: Allows the user to select and submit a new INR value.
- INR History Screen: Displays previous INR records and trend-style visualization.
- User Profile Screen: Allows the user to edit treatment information.
- Settings Screen: Allows light/dark theme selection and language preference.
- Medical Warnings Screen: Presents general safety warnings.
- Food Interactions Screen: Presents vitamin K related food information.
- Drug Interactions Screen: Presents Warfarin-related drug interaction information.
- Privacy Policy Screen: Explains data and privacy handling.
- About Screen: Explains the project.
- Help Screen: Provides user guidance.
- Delete Account Screen: Handles account deletion separately from normal profile editing.

The UI was designed with elderly and non-technical users in mind. The design uses readable text, simple cards, limited button count, clear navigation, and high-contrast elements. Tablet icons and half-tablet visuals were added so that the daily medication plan is not only written as text but also represented visually.

Many detailed UI refinements were made during development. Card alignment, color palette, button size, spacing, typography, responsive layout, weekly calendar placement, information icons, daily dose card, warning cards, drawer menu behavior, and back button placement were adjusted repeatedly. A right-side drawer menu was added on the home screen for navigation. The settings and theme system were also expanded so that light and dark modes apply consistently across the application.

One important design improvement was the separation of destructive account deletion from the normal profile screen. Instead of placing the full deletion process directly inside profile editing, the profile screen contains a clear button that navigates to a dedicated Delete Account screen. This screen asks for password reauthentication and confirmation before deleting the account and related Firestore data.

Backend and Database Development

The backend of the project is Firebase-based. There is no custom Node.js/Express backend server in the current implementation. Node.js is used only as part of the development toolchain for packages, Expo, EAS, TypeScript, and tests. The actual backend responsibilities are handled by Firebase Authentication, Cloud Firestore, and Firestore Security Rules.

Firebase Authentication manages:

- User registration.
- User login.
- Session persistence.
- Logout.
- Password-based reauthentication before account deletion.
- Final Firebase Auth user deletion.

Cloud Firestore manages:

- User profile documents.
- Treatment indication.
- Target INR values.
- Current weekly dose.
- Theme preference.
- Language preference.
- INR measurement records.
- Recommendation-related data.

The main Firestore-related service functions include creating a user profile if missing, updating user profile data, retrieving the user profile, adding INR records, retrieving the latest INR record, retrieving all INR records, updating theme mode, updating language, and deleting user data. Account deletion uses batch operations to remove the user's INR records and recommendation documents before deleting the user profile document.

The main user data model includes fields such as `uid`, `email`, `createdAt`, `indication`, `targetInrMin`, `targetInrMax`, `currentWeeklyDoseMg`, `themeMode`, `language`, and `requiresInitialIndication`. INR record documents include `uid`, `inr`, `measuredAt`, and `createdAt`. Recommendation structures include input values, algorithm output, warning objects, action summary, daily plan, and creation time.

Using Firebase provided several advantages for this project. It reduced backend development time, allowed authentication and database features to be integrated quickly, and provided a scalable serverless infrastructure. It also made the app suitable for an academic prototype because the development effort could focus on clinical logic, UI, and user safety instead of building a custom server from scratch.

Security, Privacy, and Medical Safety

Because the application stores health-related information, security and privacy were treated as important project requirements. Firebase Authentication ensures that each user has a unique identity. Firestore documents store the user's `uid`, and security rules compare the authenticated user ID with the document owner ID.

The Firestore rules follow an ownership model:

- A user can read and delete only their own user document.
- A user can create or update a user document only when the incoming data belongs to their own UID.
- INR records can be created only when the incoming record UID matches the authenticated user.
- INR records and recommendation documents can be read, updated, or deleted only by their owner.
- All other document access is denied by default.

This rule structure prevents one user from reading or modifying another user's INR records or profile information. It also makes the data model easier to explain during a technical review because the `uid` is the link between authentication and stored data ownership.

Medical safety was handled through disclaimers, consent, warning messages, and conservative rule design. The application clearly presents itself as an academic prototype and decision-support concept. It does not replace physician evaluation, diagnosis, or emergency care. High-risk INR values generate stop/recheck recommendations and warning codes rather than simple dose changes. The application also includes separate screens for medical warnings, food interactions, drug interactions, privacy policy, and help.

Food and Drug Interaction Module

Food and drug interactions are important in Warfarin therapy because vitamin K intake and medication interactions can change the anticoagulant effect. A food interaction list was prepared with items such as cabbage, spinach, parsley, broccoli, Brussels sprouts, asparagus, cauliflower, avocado, green tea, liver, and other foods grouped by vitamin K relevance. The main message is not always to completely avoid these foods, but to keep intake consistent and follow the physician's advice.

A drug interaction information module was also prepared. The research stage included antibiotics, amiodarone, paracetamol, corticosteroids, alcohol, birth control medications, carbamazepine, antihistamines, and other substances that may increase or decrease Warfarin's effect. In the current application, these lists are mainly educational. They are not yet fully integrated into the dose algorithm as dynamic inputs. This was intentionally left as a future phase because algorithmic integration of interactions requires careful clinical validation.

Testing and Validation

Testing was performed at both the algorithm and application levels. Algorithm scenarios were checked using different INR values such as 1.2, 1.8, 2.4, 3.4, 4.2, 5.8, 6.8, and very high INR values. These scenarios helped verify whether low INR values triggered a dose increase, in-range values preserved the same dose, moderately high values triggered a dose decrease, and very high values triggered medication stop and short-term recheck logic.

The repository includes Jest-based test support for algorithm modules. The TypeScript structure also helps catch type errors in navigation parameters, model structures, and service functions. During development, local validation included checking JSON configuration files such as `app.json` and `eas.json`, verifying Firestore service flow, and reviewing screen behavior across user flows.

Usability testing was also considered as part of the academic evaluation. A user survey was planned for test users who try the application and provide feedback. The survey is intended to evaluate whether the application is easy to use, whether INR entry is understandable, whether dose recommendation and warning messages are clear, whether the visual tablet schedule is helpful, whether the design is suitable for the target user group, and whether users feel guided while using the app. A reasonable survey scope is 12-15 questions covering usability, clarity, safety perception, design, and overall satisfaction.

Google Play Release Preparation

The project also included a detailed Android release preparation process. The application was prepared for Google Play under the Android package name `com.yesimarslan.inrmonitoring`. The Expo configuration includes version `1.0.2`, Android `versionCode` 3, app icon, splash screen, adaptive icon foreground, adaptive icon background, monochrome icon, and EAS project ID.

The release workflow used Expo Application Services (EAS). The `production` build profile in `eas.json` produces an Android App Bundle (`.aab`) for Google Play upload. A separate `preview` profile was added to produce an APK file for direct sharing with the instructor through Google Drive. This distinction was important because `.aab` is the correct format for Play Console, while `.apk` is easier for direct installation and testing outside Play Console.

The Google Play preparation process included:

- Application name preparation.
- App icon preparation.
- Adaptive icon preparation.
- Splash screen preparation.
- Feature graphic preparation.
- Phone screenshots.
- 7-inch tablet screenshots.
- 10-inch tablet screenshots.
- Privacy policy preparation.
- Medical disclaimer preparation.
- Firebase security rules preparation.
- Package name configuration.
- Android versionCode management.
- EAS build configuration.
- Internal/closed/open testing workflow research.
- APK sharing workflow for instructor review.

During the Play Console process, an important rejection/problem was encountered: Play Console did not accept a new upload when the Android `versionCode` was the same as the previous uploaded bundle. This was resolved by increasing `android.versionCode` in `app.json`. The first recorded release fix increased versionCode from 1 to 2 and the visible version from 1.0.0 to 1.0.1. Later, another release preparation step increased versionCode from 2 to 3 and version to 1.0.2. This showed that Play Console treats `versionCode` as the main technical gate for replacement uploads.

Another Play Console warning was related to a missing mapping/deobfuscation file. This warning affects how readable crash reports are if code obfuscation is used, but it was identified as non-blocking by itself. The more important rejection risks for a medical-related application are privacy policy problems, incorrect data safety forms, missing medical disclaimers, crashes, test account issues, or incomplete store listing information.

Google Play testing tracks were also reviewed. Google Play does not use Apple TestFlight; Android testing is handled through internal, closed, and open testing tracks. Internal testing is suitable for a small number of trusted testers and quick QA. Closed testing is suitable for a wider but controlled tester group. Open testing can make the test version visible to a broader group. For this project, the practical path was to prepare EAS builds, upload the `.aab` to Play Console, configure testing, and share test links or APK files depending on the evaluation need.

Academic Project Process

The project was not only a software implementation but also an academic research and documentation process. In the early stage, the problem was defined around the difficulties of Warfarin management and INR follow-up. A literature review was performed, similar applications were examined, and the project scope was expanded from simple tracking to rule-based clinical decision support.

The project was later converted into a TUBITAK 2209-B project proposal. The proposal included the problem definition, literature review, innovative contribution, technical architecture, clinical algorithm, frontend/backend design, risk management, research schedule, commercial potential, and scientific contribution. A detailed project report of approximately 20 pages was prepared, and the project was accepted for support.

Figma designs were created before and during development. These included login, registration, home, profile, INR entry, weekly calendar, menu, drug lists, food lists, and help screens. The design principles were simplicity, large readable text, high contrast, minimal unnecessary buttons, and suitability for elderly users.

The project was also prepared for academic presentations. Introductory, intermediate, and final presentations were produced. Demo videos and presentation scripts were prepared in English. An academic poster was created with problem definition, solution, algorithm, technologies, screenshots, QR code, GitHub link, and Linktree information. The project repository was documented with README, architecture notes, algorithm notes, roadmap, and supporting documents.

Results

At the end of the project, a working mobile application prototype was developed. The application supports user registration, login, medical consent, initial indication selection, profile management, INR entry, latest INR display, dose recommendation generation, weekly dose visualization, INR history, medical warnings, food interactions, drug interactions, help, about, privacy policy, settings, theme/language preferences, logout, and account deletion.

The clinical algorithm can generate different outputs depending on INR and indication. It can increase, decrease, or preserve the weekly dose; recommend temporary medication stop in high INR scenarios; generate warning codes; and provide next INR control timing. The weekly dose is rounded to practical tablet steps and displayed as a daily plan.

The backend successfully integrates Firebase Authentication and Cloud Firestore. User data is organized with clear model structures, and Firestore Security Rules protect ownership. Account deletion removes related Firestore data and then deletes the Firebase Authentication user after password reauthentication.

The release side also reached a mature preparation stage. The project includes Android package configuration, EAS production and preview build profiles, versionCode handling, Play Console asset preparation, privacy policy, medical disclaimer, and APK sharing workflow.

Overall, the project achieved its main goal: it demonstrates a personalized INR monitoring and Warfarin dose optimization system as an academic clinical decision-support prototype.

Limitations

The project has several limitations. First, it is an academic prototype and not a certified medical device. It must not be used as a replacement for physician decisions. Second, the rule-based dose algorithm was implemented from clinical tables and advisor guidance, but it still requires broader clinical validation before real-world medical use. Third, the interaction module is currently educational and is not yet fully integrated into the algorithm as a dynamic adjustment input. Fourth, the current backend does not include a doctor/admin panel. Fifth, the machine learning phase has not yet been implemented because patient-specific learning requires sufficient validated data.

Another limitation is that the active dose calculation currently runs in the mobile application layer. This is acceptable for an academic prototype, but a production medical system would ideally place validated decision logic behind a controlled backend or cloud function layer with audit logging, versioning, and clinical approval workflow.

Future Work

Future work can improve the project in several directions. The priority is stronger clinical validation with physician review and more test scenarios. The warning system can be expanded, and the food/drug interaction module can be structured as a more formal data model. A doctor-facing dashboard can be added so that physicians can review patient INR trends and generate recommendations. Exportable PDF reports can be generated for clinic visits.

The second major future direction is machine learning-based personalization. After collecting enough validated patient data, the system may learn patient-specific INR response patterns. Possible ML tasks include predicting the next INR value, predicting whether a patient will be inside the target range, detecting unstable INR patterns, and recommending smaller personalized dose adjustments. However, the ML model should not override safety constraints. A safer architecture would allow the ML model to suggest a recommendation while a rule-based safety validator enforces hard clinical boundaries.

Other future improvements include offline mode, push notifications for INR control dates, medication reminders, better INR trend charts, multilingual expansion, doctor-patient communication, improved accessibility, and wider testing through Google Play closed testing.

Conclusion

This project developed a personalized INR monitoring and Warfarin dose optimization mobile application prototype. It began as a simple INR tracking idea but evolved into a broader clinical decision-support system with rule-based dose adjustment, weekly tablet scheduling, warning messages, interaction education, Firebase backend integration, security rules, academic documentation, and Google Play release preparation.

The project shows how software engineering can support a real healthcare workflow when clinical safety, explainability, usability, and data privacy are considered together. The final system is not a replacement for medical care, but it demonstrates how mobile health tools can help patients better understand INR values, medication plans, warnings, and follow-up timing. With further clinical validation, backend strengthening, and future personalization features, the system can become a stronger digital health solution for Warfarin management.

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